Notes: Tombe and Zhu (AER, 2019)

Trade, Migration, and Productivity: A Quantitative Analysis of China

Contents

1 Big picture

- **Question.** How much of China's rapid growth in the early 2000s came from falling *internal* trade frictions and migration frictions, as opposed to falling *external* trade frictions after WTO accession?
- **Setting.** The paper studies 30 mainland Chinese provinces (Tibet excluded), split into agriculture and non-agriculture, and combines internal trade, international trade, and worker migration across regions and sectors in one quantitative general equilibrium model.
- **Core institutional frictions.** Two distortions are central: local trade barriers and transport frictions across provinces, and the hukou plus land institutions that make it costly for workers to leave their place and sector of registration.
- **Main empirical fact.** China in 2000 had enormous real income dispersion across provinces and a very large productivity gap between agriculture and non-agriculture. Yet inter-provincial migration was still small as a share of total employment, which implies large barriers to labor reallocation.
- **Main result.** Reductions in internal trade costs and migration costs explain about 28 percent of aggregate labor-productivity growth between 2000 and 2005, while reductions in external trade costs explain only about 8 percent.
- **Why internal reforms dominate.** Most Chinese provinces trade more with the rest of China than with the rest of the world, and labor reallocation from agriculture to non-agriculture is quantitatively very important. Internal integration therefore has much larger aggregate consequences than export expansion alone.
- **Mechanism.** Lower trade costs reduce home bias and let regions specialize according to comparative advantage; lower migration costs let workers move toward higher real-income locations and, especially, from agriculture to non-agriculture.
- **Welfare result.** The welfare gains are even larger than the GDP gains because lower migration costs directly raise utility for migrants in addition to improving allocation.
- **Broader message.** The common "export-led growth" narrative misses a central part of the Chinese experience. The paper argues that domestic integration—goods-market integration, labor mobility, and land institutions—is at least as important as external liberalization.
- **Policy takeaway.** Even after the declines measured in the paper, migration and internal trade frictions remain high. Further reform, especially on migration and land institutions, has potentially very large welfare gains.

2 Data and measurement (key objects)

2.1 Spatial income data and sectoral structure

- **Unit of analysis.** The economy is divided into 30 Chinese provinces and two sectors: agriculture and non-agriculture.
- **GDP and employment.** Provincial nominal GDP and employment by sector come from the *China Statistical Yearbook*.

- **Price levels.** To construct real GDP per worker, the paper combines rural and urban CPI series with province-level rural and urban price levels from Brandt and Holz (2006), built from retail and procurement price data in the 1990s.
- **Real-income fact.** In 2000, the ratio of average real GDP per worker in the top five provinces relative to the bottom five provinces was roughly 4:1.
- **Sectoral gap.** Within a province, non-agricultural real GDP per worker was much larger than agricultural real GDP per worker; the average within-province ratio was also about 4:1.
- **Interpretation.** Large gaps across space and sectors create large potential gains from labor reallocation, provided workers can move.

2.2 Migration data and definitions

- **Main source.** The migration analysis uses the 1 percent sample of the 2000 Population Census and the 20 percent sample of the 2005 1 percent mini-census.
- **Why not household surveys?** Residence-based rural and urban household surveys understate migration, especially migrant workers without local registration.
- **Inter-provincial migration.** A worker employed in a province different from the province of hukou registration is counted as an inter-provincial migrant.
- **Intra-provincial migration.** A worker in the hukou province but employed in a different hukou category—for example, agricultural hukou and non-agricultural work—is counted as an intra-provincial migrant.
- **Key levels.** In 2000 there were 26.5 million inter-provincial migrant workers and 90.1 million intra-provincial migrant workers. By 2005 these rose to 49.0 million and 120.4 million.
- **Key shares.** Inter-provincial migrants rose from 4.2 percent to 7.2 percent of employment; intra-provincial migrants rose from 14.3 percent to 17.7 percent.
- **Composition.** Most of the increase reflects rural-to-urban migration, especially workers with agricultural hukou moving into non-agriculture.

2.3 Trade data and trade shares

- **Main source.** Internal and external trade flows are constructed from the interregional input-output tables for 2002 and 2007 by Li (2010) and Zhang and Qi (2012).
- **Coverage.** These data combine provincial tables, surveys of sources of inputs and destinations of output, and transportation information to recover bilateral trade flows.
- **Normalization.** The authors normalize bilateral trade by the importing region's total expenditure to construct expenditure shares π_{ni}^{j}.
- **Internal versus external openness.** Most Chinese regions import more from the *rest of China* than from abroad, even if imports from any single foreign or domestic partner are smaller.
- **Key pattern.** Interior regions have higher home shares than coastal regions, and home shares fall between 2002 and 2007, indicating rising openness.

- **Important fact for the paper.** On average, the increase in the import share from the rest of China is much larger than the increase in the import share from abroad, which mechanically raises the importance of internal integration for aggregate outcomes.

2.4 Institutions behind the measured frictions

- **Hukou system.** Each individual is assigned an agricultural or non-agricultural hukou in a specific place, and changing either the category or the location is difficult.
- **Flow migration cost.** Migration is modeled as an *ongoing* cost rather than a one-time sunk cost, because migrants without local hukou face continued disadvantages in access to public services, education, and labor-market rights.
- **Land institution.** Local residents share the returns to collectively owned land and structures; migrants who leave lose those returns. This is why the model gives land income only to local hukou holders in the baseline.
- **Internal trade frictions.** Internal trade barriers reflect both policy distortions (local protectionism tied to state-sector interests) and standard transport/logistics costs.
- **Policy changes around 2000.** The paper emphasizes the 2001 directive against local protectionism, SOE reforms, infrastructure improvements, and the 2003 relaxation/streamlining of temporary residence requirements for migrants.

3 Models and empirical specifications (core equations + interpretation)

3.1 A motivating back-of-the-envelope decomposition

Let y_n^j denote real GDP per worker in region n and sector j, let l_n^j be the corresponding employment share, and define aggregate labor productivity by

$$y = \sum_{n,j} y_n^j l_n^j. \tag{1}$$

Following a shock that changes regional-sectoral productivity and employment, aggregate growth can be decomposed as

$$\begin{aligned} \widehat{y} &= \sum_{n,j} \omega_n^j \widehat{y}_n^j \widehat{l}_n^j \\ &= 1 + \sum_{n,j} \omega_n^j g_{y_n^j} + \sum_{n,j} \omega_n^j g_{l_n^j} + \sum_{n,j} \omega_n^j g_{y_n^j} g_{l_n^j}, \end{aligned} \tag{2}$$

where $\omega_n^j \propto y_n^j l_n^j$ is the initial GDP weight and $g_x = \widehat{x} - 1$.

The paper then uses the home-share formula from trade theory:

$$\widehat{y}_n^j = \widehat{A}_n^j (\widehat{\pi}_{nn}^j)^{-1/\theta}, \qquad g_{y_n^j} \approx g_{A_n^j} - \frac{1}{\theta} \frac{\Delta \pi_{nn}^j}{\pi_{nn}^j}. \tag{3}$$

Writing home-share changes as the sum of a change in the share imported from the rest of China (π^j_{nc}) and a change in the share imported from abroad (π^j_{nw}), aggregate growth becomes

$$g_y = \sum_{n,j} \omega^j_n \frac{1}{\theta} \frac{\Delta \pi^j_{nc}}{\pi^j_{nn}} + \sum_{n,j} \omega^j_n \frac{1}{\theta} \frac{\Delta \pi^j_{nw}}{\pi^j_{nn}} + \sum_{n,j} \omega^j_n g_{l^j_n} + \sum_{n,j} \omega^j_n g_{A^j_n}. \tag{4}$$

- **Interpretation.** The four terms are, respectively: internal-trade gains, external-trade gains, migration/reallocation gains, and residual productivity growth.
- **Quantitative benchmark.** This simple decomposition assigns 4.9 percent of growth to internal trade, 0.5 percent to external trade, 10.8 percent to migration, and 40.9 percent to the residual.
- **Why only a benchmark?** It ignores general equilibrium interactions between trade and migration, intermediate inputs, land as a fixed factor, and changes in underlying trade and migration costs.

3.2 Economic environment and worker preferences

There are $N+1$ regions: China's N provinces plus the rest of the world. Each region has two sectors, $j \in \{\text{ag}, \text{na}\}$, and each region-sector has a fixed factor $\bar{S}^j_n$ used for both production and housing.

Workers registered in region n and sector j have Cobb-Douglas preferences over agricultural goods, non-agricultural goods, and housing:

$$u^j_n = \varepsilon^j_n \left[(C^{j,\text{ag}}_n)^{\psi^{\text{ag}}} (C^{j,\text{na}}_n)^{\psi^{\text{na}}} \right]^{\alpha} (S^{j,h}_n)^{1-\alpha}. \tag{1}$$

The resulting final demand for good j in region n is

$$D^j_n = \alpha \psi^j \sum_{k \in \{\text{ag},\text{na}\}} v^k_n L^k_n. \tag{2}$$

- **Interpretation of $(\alpha, \psi^{\text{ag}}, \psi^{\text{na}})$.** α is the expenditure share on goods rather than housing; ψ^{ag} and ψ^{na} split goods demand between agriculture and non-agriculture.
- **Role of ε^j_n.** This is an idiosyncratic location-sector preference shifter. It lets otherwise similar workers sort differently across provinces and sectors.
- **Why this matters.** The preference heterogeneity is what makes the migration share equation analytically tractable later on.

3.3 Production, trade, and goods prices

Sector-j output in region n is a CES aggregate over varieties. Perfectly competitive firms produce each variety using labor, intermediate inputs, and land. The marginal cost of producing a variety with productivity φ is

$$c^j_n(\varphi) \propto \frac{1}{\varphi} \left[(w^j_n)^{\beta^j} (r^j_n)^{\eta^j} \left(\prod_{k \in \{\text{ag},\text{na}\}} (P^k_n)^{\sigma^{jk}} \right) \right]. \tag{3}$$

With Fréchet productivity draws, bilateral trade shares are

$$\pi_{ni}^{j} = \frac{T_i^j(\tau_{ni}^j c_i^j)^{-\theta}}{\sum_{m=1}^{N+1} T_m^j(\tau_{nm}^j c_m^j)^{-\theta}}. \tag{4}$$

Sectoral price indices are

$$P_n^j = \gamma \left[\sum_{i=1}^{N+1} T_i^j(\tau_{ni}^j c_i^j)^{-\theta}\right]^{-1/\theta}. \tag{5}$$

Revenue and expenditure satisfy

$$R_n^j = \sum_{i=1}^{N+1} \pi_{in}^j X_i^j, \tag{6}$$

$$X_n^j = D_n^j + \sum_k \sigma^{kj} R_n^k. \tag{7}$$

- **θ.** This is the trade elasticity. A larger θ means trade shares react more to trade costs.
- **β^j and η^j.** These are labor and land shares in production; agriculture is much more land intensive than non-agriculture in the calibration.
- **σ^{jk}.** These input-output shares are what allow lower trade costs to affect not just final consumption but production efficiency through cheaper intermediates.
- **Central economic channel.** Trade costs matter because they alter expenditure shares and sectoral price indices, which then alter real incomes and labor allocation.

3.4 Worker income, local land ownership, and the rebate term

Because local residents collectively own land, migrants do not receive fixed-factor income in the baseline. The total fixed-factor income in region n, sector j is

$$r_n^j \bar{S}_n^j = \left[\frac{(1-\alpha)\beta^j + \eta^j}{\alpha\beta^j}\right] w_n^j L_n^j. \tag{8}$$

Define the effective fixed-factor rebate to workers as

$$\delta_{ni}^{jk} = \begin{cases} 1 + \left(\dfrac{(1-\alpha)\beta^j + \eta^j}{\alpha\beta^j}\right) \dfrac{L_n^j}{L_{nn}^{jj}} & \text{if } n = i \text{ and } j = k, \\ 1 & \text{if } n \neq i \text{ or } j \neq k. \end{cases} \tag{9}$$

Then the income of a worker registered in (n, j) but working in (i, k) is

$$v_{ni}^{jk} = \delta_{ni}^{jk} w_i^k. \tag{5}$$

- **Interpretation.** Staying local delivers a wage plus a land-income rebate; migrating delivers only the wage.
- **Why this is important.** The land institution makes migration costly even when goods-market frictions are low. It is a major reason the paper treats migration costs as more than pure disutility or transport costs.
- **Modeling contribution.** This differs from standard spatial trade models that rebate land income proportionally or lump-sum; here the Chinese institution is built into the migration wedge itself.

3.5 Migration shares and the role of κ

Let m_{ni}^{jk} be the share of workers registered in (n, j) who work in (i, k). If location preferences are Fréchet with dispersion parameter κ, then the share moving from (n, j) to (i, k) is

$$m_{ni}^{jk} = \frac{(V_i^k \delta_{ni}^{jk} / \mu_{ni}^{jk})^{\kappa}}{\sum_{k'} \sum_{i'=1}^{N} (V_{i'}^{k'} \delta_{ni'}^{jk'} / \mu_{ni'}^{jk'})^{\kappa}}, \qquad L_i^k = \sum_j \sum_{n=1}^{N} m_{ni}^{jk} \bar{L}_n^j. \tag{10}$$

Here V_i^k is the real wage in destination (i, k) and μ_{ni}^{jk} is the migration cost.

- **Interpretation.** Migration is increasing in destination real income and decreasing in migration costs.
- **Role of κ.** A larger κ means lower idiosyncratic dispersion in tastes, so workers respond more strongly to real-income differences.
- **Two types of migration cost.** The paper measures both within-province sectoral switching costs and between-province costs, and these behave very differently in the data.

3.6 Counterfactual welfare and real GDP

Using Dekle-Eaton-Kortum style "hat algebra," the paper solves directly for changes rather than levels. Aggregate welfare changes according to

$$\widehat{W} = \sum_j \sum_{n=1}^{N} \omega_n^j \widehat{V}_n^j \, \widehat{\delta}_{nn}^{jj} \, (\widehat{m}_{nn}^{jj})^{-1/\kappa}, \tag{11}$$

and real GDP changes according to

$$\widehat{Y} = \sum_j \sum_{n=1}^{N} \phi_n^j \widehat{V}_n^j \widehat{L}_n^j. \tag{12}$$

- **Interpretation.** Welfare depends both on real income and on the direct utility effect of migration costs; GDP depends on real income times employment allocation.
- **Why welfare can exceed GDP gains.** Lower migration costs directly raise utility for movers even before any productivity effect is counted.
- **Why hat algebra helps.** The quantitative exercises require only changes in trade and migration costs, so they are less sensitive to any time-invariant bias in estimated levels.

3.7 Estimating the migration elasticity

The paper estimates κ from migration shares. Under alternative assumptions about migration costs, the estimating equations are

$$\ln\left(\frac{m_{ni}^{jk}}{m_{nn}^{jj}}\right) = \kappa \ln(V_i^k) - \varrho\kappa \ln d_{ni} + \gamma_n^j + \varsigma_{ni}^{jk}, \qquad (n, i) \neq (i, k), \tag{13}$$

or

$$\ln\left(\frac{m_{ni}^{jk}}{m_{nn}^{jj}}\right) = \kappa \ln(V_i^k) + \gamma_{ni} + \gamma_n^j + \varsigma_{ni}^{jk}, \qquad (n, i) \neq (i, k). \tag{14}$$

- **Interpretation.** Conditional on origin-sector fixed effects and either distance or origin-destination fixed effects, destination real income should attract more migrants.
- **Estimated range.** Across OLS and IV specifications, the paper gets estimates between about 1.19 and 1.61 and calibrates $\kappa = 1.5$.
- **Distance effect.** Distance enters negatively, which is exactly what one would expect if between-province movement is substantially harder than within-province sector switching.

3.8 Inferring migration and trade costs

Once κ is known, migration shares imply bilateral migration costs. For trade, the paper uses the Head-Ries index:

$$\bar{\tau}_{ni}^{j} \equiv \sqrt{\tau_{ni}^{j}\tau_{in}^{j}} = \left(\frac{\pi_{nn}^{j}\pi_{ii}^{j}}{\pi_{ni}^{j}\pi_{in}^{j}}\right)^{1/(2\theta)}. \tag{15}$$

- **Why this is convenient.** The trade-cost formula uses only observed trade shares and the elasticity θ.
- **Symmetry issue.** The raw Head-Ries object is symmetric, so the paper adds exporter-specific asymmetries through a gravity-style adjustment.
- **Migration-cost finding.** Average migration cost in 2000 is 2.82, but it is much larger for between-province movement, especially for agricultural-to-non-agricultural switching across provinces.
- **Trade-cost finding.** Both internal and external trade costs fall over time, but internal trade-cost changes matter more because the underlying internal trade shares are larger.

3.9 Land reform counterfactual

To study individual ownership land reform, the paper lets workers retain land income regardless of where they live. The modified rebate term is

$$\delta_{ni}^{jk} = 1 + \frac{(1-\alpha)v_n^j L_n^j + (\eta^j/\beta^j)w_n^j L_n^j}{w_i^k \bar{L}_n^j}, \tag{16}$$

which implies, holding migration costs fixed,

$$\frac{\widehat{m}_{ni}^{jk}}{\widehat{m}_{nn}^{jj}} = \left(\frac{\widehat{\delta}_{ni}^{jk}\widehat{V}_i^k}{\widehat{\delta}_{nn}^{jj}\widehat{V}_n^j}\right)^{\kappa}. \tag{17}$$

- **Interpretation.** Once migrants keep land income from their home location, the baseline penalty from leaving disappears.
- **Key implication.** This increases migration but not necessarily measured GDP, because the reform can shift workers toward lower-wage rural areas while still increasing welfare.

4 Identification and instruments (what makes the quantitative exercise credible)

4.1 What is identified directly from data?

- **Migration shares m_{ni}^{jk}.** These come from the census and mini-census.
- **Trade shares π_{ni}^{j}.** These come from the interregional input-output tables.
- **Regional-sectoral real incomes V_n^j.** These come from real GDP per worker after adjusting for rural and urban price differences.
- **Calibration targets.** Table 3 ties down the rest of the model using expenditure shares, input-output shares, observed labor allocations, and standard elasticity values.

4.2 Structural identification of migration costs

- **Logic.** Given observed migration shares, observed income differences, and a calibrated κ, the model backs out the migration wedge μ_{ni}^{jk} required to rationalize the data.
- **Main concern.** If provinces differ systematically in amenities, then low migration from poor to rich places may reflect preferences rather than true migration costs.
- **Paper's response.** The authors explicitly acknowledge this possibility, but stress that their main quantitative exercises rely on *changes* in migration costs. A time-invariant amenity bias does not distort those change-based counterfactuals.

4.3 Identification of κ

- **Endogeneity problem.** Destination real income may be correlated with unobserved migration barriers or local institutional quality.
- **Instrument 1: neighboring income.** The distance-weighted average income of neighboring provinces predicts destination income but is plausibly orthogonal to a province's bilateral migration wedge.
- **Instrument 2: Bartik-style expected income.** The paper combines national subsector earnings with each province's subsector employment composition to form an expected-income instrument.
- **Empirical conclusion.** The destination-income coefficient is robustly positive, and the authors choose $\kappa = 1.5$ as a central estimate.

4.4 Identification of trade costs

- **Trade-cost inversion.** Trade costs are not estimated from a reduced-form policy shock; they are inverted from bilateral trade shares using the structural gravity equation.
- **Exporter asymmetry.** Because the Head-Ries index only identifies symmetric bilateral costs, exporter-specific fixed effects are used to recover directional asymmetries.

- **Interpretation.** This is a structural measurement exercise, not a natural-experiment design. Credibility comes from the discipline imposed by the model and the richness of the bilateral flow data.

4.5 Overall credibility of the decomposition

- **Strength.** The model jointly matches trade flows, migration flows, and real-income differences, which is hard to do with a partial-equilibrium approach.
- **Strength.** It explicitly incorporates Chinese institutions—hukou and local land ownership—instead of treating labor mobility as frictionless or migration costs as a generic black box.
- **Limitation.** The paper does not separately identify whether the calibrated productivity changes $\widehat{T}_n^j$ reflect pure TFP growth, capital deepening, or improved within-region capital allocation.
- **Limitation.** Since the decomposition is model-based, the estimated shares of growth due to internal trade, external trade, and migration are only as credible as the structural assumptions that map flows into costs.

5 Tables and figures

5.1 Figure 1: Spatial distribution of real incomes and migration in 2000

Read:

- The map of real GDP per worker shows a sharp east-coast advantage: Beijing, Shanghai, Guangdong, and nearby coastal provinces are much richer than interior provinces.
- The map of migrant shares shows that migrants concentrate in the same high-income coastal destinations.
- The figure visually links the paper's two central objects: strong income gradients and limited but directionally large migration toward rich provinces.
- It also motivates why hukou and land restrictions matter: if these income gaps existed in a frictionless economy, migration would likely have been much larger.

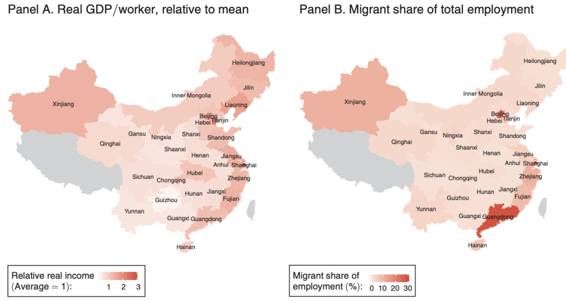

Figure 1. Spatial Distribution of Real Incomes and Migration in 2000

Notes: Displays choropleths of relative real income levels for each of China's provinces and the migrant share of total employment. Dark reds indicate both high relative real incomes and large migrant shares of employment. The gray shaded regions are Tibet, Hong Kong, and Taiwan, and are excluded from the analysis.

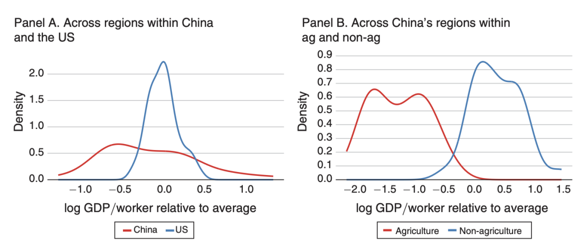

Figure 2. Spatial Distribution of GDP per Worker, 2000

Notes: Displays distribution of nominal GDP per worker across regions. Panel A compares aggregate values across China's provinces relative to the distribution across US states. Panel B displays values across regions of China within agriculture and non-agricultural sectors. Data for the United States are from the Bureau of Economic Analysis's state-level GDP and employment data. All data are for the year 2000.

5.2 Figure 2: Spatial distribution of GDP per worker in China and the US

Read:

- Panel A shows that dispersion across Chinese provinces is far wider than across US states.
- Panel B shows an even more dramatic separation between agriculture and non-agriculture inside China.
- The quantitative model is built to explain why such large productivity gaps coexist with incomplete migration.

5.3 Table 1: Stock of migrant workers in China

Read:

- In absolute terms, both inter-provincial and intra-provincial migration are huge, but intra-provincial migration is much larger throughout.
- Inter-provincial migration rose from 26.5 to 49.0 million between 2000 and 2005, while intra-provincial migration rose from 90.1 to 120.4 million.
- The most relevant rows are the employment shares: despite large growth, cross-province migration is still only 4.2 to 7.2 percent of employment.
- This table is the first hard piece of evidence that labor mobility barriers remain quantitatively large.

Table 1—Stock of Migrant Workers in China

	Inter-provincial		Intra-provincial	
	2000	2005	2000	2005
Total migrant stock (millions)	26.5	49.0	90.1	120.4
Share of total employment (%)				
Total migrants	4.2	7.2	14.3	17.7
Agriculture-to-non-agriculture migrants	3.4	5.6	13.1	16.4

Notes: Migrants are defined based on their *hukou* location. Inter-provincial migrants are workers registered in another province from where they are employed. Intra-provincial migrants are workers registered in the same province where they are employed, but are either non-agricultural workers holding agricultural *hukou* or vice versa.

Table 2—Internal and External Trade Shares of China

	Exporter									
Importer	Northeast	Beijing Tianjin	North Coast	Central Coast	South Coast	Central region	Northwest	Southwest	Abroad	Total other prov.
Year 2002										
Northeast	87.9	0.7	1.0	0.8	1.3	1.1	0.8	0.9	5.5	6.6
Beijing/Tianjin	3.9	63.4	9.4	3.0	2.6	3.3	1.4	1.2	11.9	24.8
North Coast	1.8	3.3	79.8	3.4	1.8	3.8	0.9	0.8	4.4	15.8
Central Coast	0.2	0.2	0.6	81.0	1.5	2.4	0.5	0.5	13.3	5.7
South Coast	0.5	0.4	0.5	2.6	72.3	1.9	0.4	1.5	19.8	7.9
Central region	0.6	0.3	1.1	4.8	2.3	87.8	0.7	0.7	1.8	10.4
Northwest	2.0	0.8	2.1	3.3	4.5	3.6	77.4	3.8	2.6	20.0
Southwest	0.9	0.3	0.4	1.8	4.3	1.4	0.9	88.0	2.0	10.0
Abroad	0.0	0.0	0.0	0.1	0.2	0.0	0.0	0.0	99.6	–
Year 2007										
Northeast	78.7	2.0	2.0	0.9	2.7	1.0	1.4	0.9	10.4	10.9
Beijing/Tianjin	3.8	62.3	10.1	1.5	2.4	1.8	2.1	0.7	15.5	22.2
North Coast	2.1	5.8	76.8	1.5	1.5	3.7	2.3	0.8	5.5	17.7
Central Coast	1.1	0.7	1.4	76.8	1.8	4.8	1.7	0.9	10.8	12.4
South Coast	1.5	0.9	1.7	5.2	68.5	3.6	1.8	2.8	14.1	17.4
Central region	1.7	1.4	4.5	4.9	4.0	73.0	2.9	1.8	5.9	21.1
Northwest	2.3	2.2	4.8	2.7	5.5	3.6	65.6	3.6	9.8	24.6
Southwest	1.6	1.2	1.7	1.7	8.4	1.9	3.2	73.8	6.6	19.6
Abroad	0.0	0.1	0.1	0.4	0.2	0.0	0.0	0.0	99.1	–

Notes: Displays the share of each importing region's total spending allocated to each source region. See the online Appendix for the mapping of provinces to regions. The column *Total other provs.* reports the total spending share each importing region allocated to producers in other provinces of China. The diagonal elements (the "home share" of spending), the share imported from abroad, and the share imported from other provinces will together sum to 100 percent.

5.4 Table 2: Internal and external trade shares of China

Read:

- Home shares are very high, especially in the interior, which means internal trade frictions are initially large.
- Between 2002 and 2007 home shares decline almost everywhere, indicating growing openness.

- For most regions, the import share from the rest of China rises more than the import share from abroad.
- This table is the empirical reason internal trade liberalization looms so large in the final decomposition.

5.5 Table 3: Calibrated model parameters and initial values

Read:

- Agriculture is much more land intensive than non-agriculture: $(\beta^{\mathrm{ag}}, \beta^{\mathrm{na}}) = (0.29, 0.22)$ and $(\eta^{\mathrm{ag}}, \eta^{\mathrm{na}}) = (0.28, 0.03)$.
- Intermediate-input shares are sizable, especially agriculture using non-agricultural inputs, which is why intermediate-input channels matter later for trade gains.
- The calibrated elasticities are $\theta = 4$ and $\kappa = 1.5$.
- The model is disciplined directly by data on trade shares, migration shares, and hukou registrations.

TABLE 3—CALIBRATED MODEL PARAMETERS AND INITIAL VALUES

Parameter	Value	Description
(β^{ag}, β^{na})	$(0.29, 0.22)$	Labor's share of output
(η^{ag}, η^{na})	$(0.28, 0.03)$	Land's share of output
$(\sigma^{ag,na}, \sigma^{na,ag})$	$(0.60, 0.06)$	Intermediate input shares
ψ^{ag}	0.095	Agriculture's share of final demand
α	0.87	Goods' expenditure share
θ	4	Elasticity of trade
κ	1.5	Elasticity of migration
π^j_{ni}	Data	Bilateral trade shares
m^j_{ni}	Data	Bilateral migration shares
$\bar{L}^j_n$	Data	Hukou registrations

TABLE 4—THE INCOME ELASTICITY OF MIGRATION IN CHINA

	OLS		IV: Neighboring income		IV: Expected income 2005 census data		IV: Expected income 2000 UHS data	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Destination real income V^k_i	1.32	1.33	1.40	1.40	1.20	1.19	1.61	1.28
	[0.02]	[0.02]	[0.03]	[0.02]	[0.03]	[0.02]	[0.08]	[0.03]
Distance	−1.50		−1.51		−1.39		−1.51	
	[0.04]		[0.04]		[0.04]		[0.06]	
Origin-dest. prov. FEs	No	Yes	No	Yes	No	Yes	No	Yes
Origin prov.-sector FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,480	3,540	3,480	3,540	3,480	3,540	1,972	2,006
R^2	0.605	0.851	0.607	0.852	0.604	0.844	0.616	0.847
First-stage								
Neighboring income			1.02	1.02				
			[0.01]	[0.005]				
Expected income, 2005					1.12	1.11		
					[0.01]	[0.005]		
Expected income, 2000							5.29	11.69
							[0.30]	[0.30]
F-statistic			167	59	198	72	5	11

Notes: Displays the results of various regressions to estimate the income elasticity of migration. The first IV uses the distance-weighted average income of all other provinces. The second IV is a Bartik-style instrument of expected incomes using only variation in within-non-agricultural and within-agriculture composition of employment and national average earnings by subsector. For 2000, we use a subset of provinces for which the Urban Household Survey provides income data by sector. Standard errors in brackets.

5.6 Table 4: The income elasticity of migration in China

Read:

- Across OLS and IV specifications, the destination-income coefficient lies between 1.19 and 1.61.
- Distance enters negatively when included explicitly, around -1.5, which confirms the importance of geographic separation for migration barriers.
- The IV estimates remain positive and reasonably stable, which gives some comfort that the migration elasticity is not purely reflecting omitted local factors.
- The paper's choice of $\kappa = 1.5$ sits squarely in the middle of the estimated range.

5.7 Table 5: Migration rates and average migration costs

Read:

- Overall migration costs are high: the migration-weighted average cost falls from 2.82 in 2000 to 2.31 in 2005.
- The most striking entry is agricultural-to-non-agricultural migration across provinces: its average cost falls from 25.21 to 15.43, still extremely high even after reform.
- Within-province agriculture-to-non-agriculture migration is much cheaper, falling from 2.21 to 1.83.
- The table supports the paper's central claim that migration barriers are large, declining, and especially important for cross-province rural-to-urban movement.

TABLE 5—MIGRATION RATES AND AVERAGE MIGRATION COSTS

	Initial share of employment	Average migration costs μ_{in}^{kj}: Level in 2000	Level in 2005	Relative change
Overall	0.174	2.82	2.31	0.82
Agriculture to non-agriculture migration cost changes				
Overall	0.16	2.63	2.16	0.82
Within province	0.13	2.21	1.83	0.83
Between provinces	0.03	25.21	15.43	0.61
Between provinces migration cost changes				
Overall	0.04	24.75	15.08	0.61
Within agriculture	0.003	47.67	42.22	0.89
Within non-agriculture	0.01	21.02	12.2	0.58

Notes: Displays migration-weighted harmonic means of migration costs in 2000 and 2005. The migrant share of employment summarizes m_{ni}^{jk} in 2000. We use initial period weights to average the 2005 costs to capture only changes in costs and not migration patterns.

TABLE 6—BILATERAL TRADE COSTS IN 2002 AND THE CHANGE TO 2007

	Exporter								
Importer	Northeast	Beijing Tianjin	North Coast	Central Coast	South Coast	Central region	Northwest	Southwest	Abroad
Average trade cost levels in 2002									
Northeast		2.61	2.89	3.65	2.71	3.32	2.57	3.36	3.43
Beijing/Tianjin	2.60		1.92	3.13	2.44	3.08	2.66	3.44	2.84
North Coast	2.79	1.87		2.69	2.51	2.58	2.53	3.61	3.30
Central Coast	3.80	3.27	2.89		2.21	2.27	2.70	3.34	2.43
South Coast	3.74	3.39	3.59	2.91		3.03	3.08	2.93	2.62
Central region	3.18	2.94	2.53	2.15	2.06		2.46	3.12	4.08
Northwest	3.02	3.07	2.96	2.94	2.50	2.95		2.89	4.61
Southwest	3.10	3.20	3.47	2.95	1.96	3.08	2.38		4.25
Abroad	4.94	4.10	4.75	3.37	2.63	6.05	5.79	6.32	
Average trade cost changes from 2002 to 2007									
Northeast		0.91	0.90	0.84	0.83	0.88	0.92	0.88	0.81
Beijing/Tianjin	0.84		0.89	0.91	0.89	0.80	0.75	0.85	0.78
North Coast	0.87	0.93		1.00	0.87	0.78	0.72	0.77	0.80
Central Coast	0.76	0.88	0.92		0.88	0.82	0.74	0.85	0.81
South Coast	0.77	0.93	0.87	0.92		0.81	0.72	0.80	0.92
Central region	0.88	0.91	0.85	0.96	0.90		0.78	0.84	0.75
Northwest	0.99	0.92	0.85	0.96	0.88	0.86		0.87	0.68
Southwest	0.89	0.94	0.83	0.97	0.85	0.82	0.78		0.74
Abroad	0.88	0.93	0.92	0.98	1.05	0.77	0.64	0.79	

Notes: Displays the aggregate average trade costs in 2002 and the relative changes from 2002 to 2007. We aggregate the sectoral trade costs using expenditure weights, but use the sector-specific estimates in the quantitative analysis.

5.8 Table 6: Bilateral trade costs in 2002 and the change to 2007

Read:

- Bilateral trade costs are large in levels, with especially high costs involving poorer interior regions and foreign trade for many provinces.
- Most relative changes are below one, so trade costs are falling between 2002 and 2007.
- The average declines are meaningful both internally and externally, but what matters for aggregate outcomes is that internal trade starts from much larger expenditure weights.
- This table is the core measurement input for the trade counterfactuals in Section IV.

5.9 Table 7: Effects of various migration cost changes

Read:

- Using all measured migration-cost changes, within-province migrant stocks rise by 14.5 percent and between-province migrant stocks by 80.8 percent.
- Real GDP per worker rises by 4.8 percent, while welfare rises by 11.1 percent.

- Most of the gains come from lowering agriculture-to-non-agriculture migration costs; within-sector between-province migration matters much less.
- The comparison rows without land inputs, without housing, and with $\theta \to \infty$ show why the full model produces smaller migration gains than the back-of-the-envelope calculation: land and comparative advantage create diminishing returns to reallocation.

TABLE 7—EFFECTS OF VARIOUS MIGRATION COST CHANGES

	Trade shares (p.p. change)		Migrant stock (%)		Real GDP per worker (%)	Aggregate welfare (%)
	Internal	External	Within province	Between province		
All	0.1	0.1	14.5	80.8	4.8	11.1
No land inputs	0.1	0.2	14.4	85.6	5.3	8.4
And no housing	0.1	0.2	13.8	90.4	6.5	7.6
And $\theta \to \infty$	−0.2	0.1	23.2	119.2	11.8	6.2
Agriculture to non-agriculture migration cost changes						
Overall	0.1	0.1	15.2	52.9	4.3	9.1
Within provinces	−0.0	−0.1	22.8	−9.7	2.0	5.9
Between provinces	0.1	0.2	−7.0	69.9	2.8	3.5
Between provinces migration cost changes						
Overall	0.2	0.3	−7.8	97.9	3.2	5.5
Within agriculture	−0.0	0.0	−0.1	2.3	−0.0	0.1
Within non-agriculture	0.1	0.1	−1.0	30.9	0.7	2.2

Notes: Displays aggregate response to various migration cost changes. All use migration cost changes as measured, though set $\hat{\mu}_{ni}^{kj} = 1$ for certain (n, i, j, k) depending on the experiment. The change in internal and external trade shares are the expenditure weighted average changes in region's $\sum_{n \neq i} \pi_{ni}^{j}$ and $\pi_{n(N=1)}^{j}$. The migrant stock is the number of workers living outside their province of registration.

TABLE 8—EFFECTS OF TRADE COST CHANGES

	Trade shares (p.p. change)		Migrant stock (%)		Real GDP per worker (%)	Aggregate welfare (%)
	Internal	External	Within province	Between province		
Internal trade	9.2	−0.7	0.8	−1.8	11.2	11.4
External trade	−0.7	3.9	1.8	2.4	4.0	2.9
All trade	8.2	2.8	2.5	0.5	15.2	14.1
No Change in migration						
Internal trade	9.1	−0.7	–	–	11.2	11.2
External trade	−0.7	3.9	–	–	3.4	3.4
All trade	8.2	2.8	–	–	14.5	14.5
No intermediate inputs						
Internal trade	8.6	−0.5	0.3	−1.4	3.0	3.3
External trade	−0.7	3.9	1.5	1.6	1.1	0.3
All trade	7.6	3.2	1.6	0.1	4.1	3.5
No intermediate inputs and no change in migration						
Internal trade	8.6	−0.5	–	–	3.1	3.1
External trade	−0.7	3.9	–	–	0.6	0.6
All trade	7.6	3.2	–	–	3.7	3.7

Notes: Displays aggregate response to various trade cost changes. All use trade cost changes as measured, though set $\hat{\tau}_{ni}^{j} = 1$ for certain (n, i, j) depending on the experiment. The change in internal and external trade shares are the expenditure weighted average changes in region's $\sum_{n \neq i} \pi_{ni}^{j}$ and $\pi_{n(N+1)}^{j}$. The migrant stock is the number of workers living outside their province of registration.

5.10 Table 8: Effects of trade cost changes

Read:

- Lower internal trade costs raise the internal import share by 9.2 percentage points and increase real GDP per worker by 11.2 percent and welfare by 11.4 percent.
- Lower external trade costs raise GDP by only 4.0 percent and welfare by only 2.9 percent.
- Migration responses to trade-cost changes are small; internal trade liberalization slightly reduces between-province migration because poorer interior regions become more attractive in real terms.
- The bottom panels show that intermediate inputs, not migration responses, are the main reason the full model generates larger trade gains than the simple decomposition.

5.11 Table 9: Decomposing China's aggregate labor productivity growth

Read:

- Overall labor-productivity growth is 57.1 percent in continuously compounded terms.
- Productivity changes account for 36.9 points, internal trade cost changes 10.2 points, external trade cost changes 4.5 points, and migration cost changes 5.6 points.
- Thus internal trade plus migration account for 15.8 percentage points, versus only 4.5 from external trade.
- Inside migration, the quantitatively important piece is agriculture-to-non-agriculture movement, especially across provinces.

Table 9—Decomposing China's Aggregate Labor Productivity Growth

	Marginal effects		
	Real GDP per worker growth (%)	Share of growth	Standard deviation (%)
Overall (all changes)	57.1	–	–
Productivity changes	36.9	0.64	1.3
Internal trade cost changes	10.2	0.18	0.3
External trade cost changes	4.5	0.08	0.7
Migration cost changes	5.6	0.10	0.9
Of the migration cost changes			
Between-province, within-non-agriculture	0.9	0.02	0.4
Between-province, within-agriculture	0.0	0.00	0.0
Between-province, agriculture-non-agriculture	3.2	0.06	0.9
Within-province, agriculture-non-agriculture	1.5	0.03	0.3

Notes: Decomposes the change in real GDP into contributions from productivity, internal trade cost changes, external trade cost changes, and migration cost changes. The bottom panel decomposes the change due to migration cost changes into various different types of migration. To attribute contributions from each component, we report the marginal contribution to aggregate growth of each component across all permutations. In the last column, we report the standard deviation of those growth rates across permutations. Shares may not sum to 1 due to rounding. The growth rates are continuously compounded rates.

Table 10—Potential Gains from Further Trade and Migration Liberalization

	Relative to 2005 equilibrium	
	Change in real GDP (%)	Aggregate welfare (%)
Average internal trade costs as in Canada	12.5	16.3
A 1/3 inter-provincial migrant share	12.8	45.6
Both together	26.0	69.2

Notes: Reports the change in real GDP and welfare that result from changing China's internal trade and migration costs such that average trade costs correspond to estimates for Canada, and one-third of workers are outside their province of registration (a similar share as the United States). Percentage changes are expressed relative to the 2005 equilibrium.

5.12 Table 10: Potential gains from further trade and migration liberalization

Read:

- If China lowered internal trade costs to Canadian levels, real GDP would rise by 12.5 percent and welfare by 16.3 percent relative to the 2005 equilibrium.
- If China lowered migration costs enough to reach a one-third inter-provincial migrant share, real GDP would rise by 12.8 percent and welfare by 45.6 percent.
- Doing both at once raises GDP by 26.0 percent and welfare by 69.2 percent.
- The welfare gains from migration reform are especially large because migrants value the direct reduction in the migration wedge, not only the reallocation effect.

5.13 Table 11: Effect of individual ownership land reform

Read:

- Land reform raises welfare by 11.8 percent but lowers real GDP by 2.4 percent.
- Within-province migration nearly doubles and between-province migration rises by 38 percent.
- The share of agricultural workers rises from 52.9 to 56.6 percent, and urban-to-rural migration rises strongly.
- This is a subtle but important result: freeing land income from local residence improves welfare by letting people live where they prefer, even though measured output falls because more workers

Table 11—Effect of Individual Ownership Land Reform

	Percent change	
Welfare	11.8	
Real GDP	−2.4	
Migration, within-province	96.3	
Migration, between-province	38.0	
	Share of population (%)	
	Initial equilibrium	New equilibrium
Agricultural workers	52.9	56.6
Stock of migrants, urban-rural	2.1	10.9
Stock of migrants, rural-urban	14.0	19.1

Notes: Reports the change in various outcomes that results from a counterfactual where workers are permitted to keep land income, regardless of where they live. All workers registered in (n,j) receive an equal per capita land rebate $r_n^j S_n^j / \bar{L}_n^j$, even if they move to another region.

shift toward low-productivity rural locations.

6 Short summary

- **Core conclusion.** China's early-2000s growth was not mainly about export expansion; domestic integration in goods and labor markets played a larger role.
- **Central mechanism.** Lower internal trade barriers increased interprovincial trade and lowered price dispersion, while lower migration barriers moved workers out of agriculture and toward richer provinces and sectors.
- **Main quantitative lesson.** Internal trade and migration cost reductions together explain roughly 28 percent of labor-productivity growth between 2000 and 2005, compared with only 8 percent from lower external trade costs.
- **Broader lesson.** In developing economies, internal distortions—not just external openness—can be a first-order source of aggregate misallocation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Declines in internal trade and migration frictions materially raised China's aggregate labor productivity and welfare during the early reform period.
- **Relative importance.** Internal integration matters much more than external integration in the paper's accounting: 28 percent of growth versus 8 percent.
- **Magnitude.** Measured migration-cost reductions alone raise GDP per worker by 4.8 percent and welfare by 11.1 percent; measured trade-cost reductions raise GDP by 15.2 percent and welfare by 14.1 percent when internal and external trade frictions fall together.
- **Mechanism evidence.** The data show large income gaps, limited but rapidly growing migration, high home bias in trade, and strong declines in both migration and trade costs after policy changes.
- **Counterfactual implication.** There remain very large gains from further reform, especially on migration and land institutions.

7.2 Economic intuition

- China's economy in 2000 had two layers of misallocation: goods were too hard to move across provinces, and workers were too hard to move across provinces and sectors.
- Lower internal trade costs raise real income not only by allowing final-goods arbitrage, but also by making intermediate inputs cheaper where they are used in production.
- Lower migration costs matter mainly because they move labor out of low-productivity agriculture and toward high-productivity non-agriculture.

- The land institution is a genuine equilibrium wedge: it discourages migration by making movers give up locally shared rents.
- Because most provinces are much more exposed to domestic than foreign trade, domestic integration naturally has larger aggregate consequences.

7.3 Takeaway

This paper is a strong example of how to combine spatial trade theory, migration theory, and detailed institutional knowledge in a single quantitative framework. Its main punchline is simple but important: to understand China's growth, one must look inside China, not only at China's integration with the rest of the world. The paper also leaves a durable lesson for development and macro: internal distortions in labor and goods markets can be as important as, or more important than, international trade barriers in shaping aggregate productivity.